

Summary of Revisions (Round 6)

Manuscript: A Reproducible Benchmark of Relational Database Access-Layer Performance in Express.js: A Configuration-Specific Comparison on PostgreSQL and MySQL **Author:** Mateusz Miotk (University of Gdańsk) **Journal:** Information and Software Technology

I thank the reviewer for a sixth careful reading and for the clear framing that the dataset and artifact are publishable, the remaining obstacle being interpretation that claimed more causal specificity and category generality than a single-host, default-configuration design can establish. I have made that framing the organizing principle of this revision. A full point-by-point reply (every P1–P10 item, the §7 minors, the §8 statistical points, and questions Q1–Q8, each with a quotation of the revised text) is provided in the separate Response to Reviewers. The essential changes are summarized below.

Governing change: the same-SQL causal claim is withdrawn

The earlier text read the same-SQL control as a causal decomposition — that the eager-loading *strategy*, not the execution machinery, drove most of the idiomatic spread. Because collapsing every layer onto identical SQL changes query strategy, statement count, prepared-statement behaviour, the raw-versus-ORM API, and result marshalling together, the design cannot support that attribution. I have **retracted it everywhere** (both abstracts, Results, Discussion, Conclusion) and reworded the control as a pure compound **bounding contrast**: it bounds the combined query-strategy-and-formulation contribution without isolating any single mechanism. The one remaining mechanistic remark (Prisma’s slow raw path) is now labelled an explicit conjecture the contrast cannot confirm.

New experiment: longer-window tail-latency validation

Addressing the concern that the primary p99 rests on ~60 tail observations per run, I added a 60-second re-measurement of the deep fetch on both engines (five times the requests, ~300 tail observations per run) at the same operating point. The p99 ranking is **preserved exactly (Spearman rho = 1.00 per engine)**, each cell’s p99 barely moves from the 12-second value, the p97.5 ordering matches the p99 ordering, and the run-level p99 coefficient of variation stays at most 6.0% (PostgreSQL) / 3.8% (MySQL). The 12-second window is therefore adequate for the tail ranking (new Supplement Table S23).

Interpretation and framing

- **Estimand renamed.** "Idiomatic" is now a *documented default configuration under a reproducible adapter-selection rule*, explicitly not a claim of performance-optimal or typical production use.
- **Three quantities separated.** RQ1 keeps capacity, overload latency (saturated p99 at equal demand), and matched-utilization latency distinct.
- **Category claims scoped.** Practical guidance now applies to the tested implementations and configuration, offered as a hypothesis rather than a category law.
- **Perishability tempered** from a general law to one observed version-sensitivity example; the historical Prisma narrative is condensed to a single paragraph.

Statistics and reporting

- Kendall's tau added alongside Spearman's rho; "read rankings transfer" softened to "the read ordering is similar across engines."
 - Post-hoc adjacent-rank comparisons reported descriptively, not as ranking confirmation.
 - A layer-by-engine **interaction-magnitude** table (PostgreSQL÷MySQL ratios with bootstrap intervals) reports effect size rather than a p-value floor.
 - An **ex-MikroORM robustness check**: the read ordering and the deep-fetch spread (3.7x) survive excluding the largest outlier.
 - Raw per-replicate points for the noisiest (MySQL insert) cells shown as a figure.
 - The TOST margin is motivated by the engineering consequence, not by measurement noise.
 - "Intrinsic latency" replaced by capacity-normalized / matched-utilization wording;
- "CRUD-spanning" replaced by "five representative access patterns"; "native leads all ten engine-pattern combinations" stated exactly; "latest stable" dated to the freeze.

Reproducibility and length

- A new top-level **REPRODUCE.md** gives one-command smoke, full-matrix, and clean-room-from-Zenodo-tarball recipes; the archive is a git archive containing all 33 raw-data JSON files with a regenerated checksum manifest.
- The manuscript is 14,936 words (references and floats included, under 15,000); the structured abstract is 298 words. New analyses are placed in the supplement; the condensed Prisma narrative and removed repetition offset the additions.

Pre-submission consistency pass. Before submission the artefact was audited end to end so the main paper, supplement, and Zenodo archive report one version and one set of numbers. Three engine-specific tables (the deep-fetch significance table, the CPU-efficiency table S5, and the resource-footprint table S10) had been rendered for PostgreSQL while the text cites the MySQL figures; they were regenerated for MySQL to match. The relaxed-durability secondary (S3) was re-run because its archived raw file was corrupt (it showed relaxed durability slower than default); the fresh, reproducible run confirms the published result (mysql2 +2.36x on the MySQL insert) and corrects a single MikroORM outlier. The Elsevier Highlights were refreshed to current terminology and numbers, and three small cross-references were corrected. Every table now reproduces from the archived raw data (the checksum manifest verifies 33/33).

The replication package for this revision is archived at Zenodo, DOI 10.5281/zenodo.21440942 (release v1.6.3). An independent adversarial re-review reproduced every headline number from the raw data and returned a minor-revision verdict.